Where Two Panoramas Can and Cannot See Depth: Epipole-Aware Fusion for Metric 3D from Sparse Panoramas

Anonymous CVPR submission

Abstract

Panoramic depth foundation models predict a depth for every viewing direction, but their metric error of several percent is far from the centimetre accuracy of LiDAR. Two panoramas captured a few metres apart can triangulate much more precisely—yet only in some directions: along the baseline, which for street-level capture is exactly the direction of travel, depth becomes unobservable. We make this observability explicit. We derive a closed-form, pose-aware expression for the depth uncertainty of spherical two-view triangulation for every ray, which diverges at the epipoles and accounts for matching noise and relative-pose error. Combined with a two-parameter geometry-conditioned calibration of the photometric matching precision, this yields calibrated per-ray stereo uncertainties that we fuse with a learned panoramic prior in log-depth, after aligning the prior's metric scale to well-conditioned rays and gating occlusions with a mixture model. The result is metric depth with a calibrated uncertainty on every ray, degrading gracefully to the prior where geometry offers no information. We further describe how a dense city capture can act as its own teacher, supplying uncertainty-weighted and amodal supervision as well as the calibration data. On a procedural street benchmark with exact ground truth, our fusion reduces the median relative error of a simulated foundation prior from 6.95% to 1.61% with calibrated uncertainties. A geometry-blind confidence-based uncertainty fitted on the same data is equally accurate in-distribution. The closed form pays off where match confidence cannot see the geometry. Under relative-pose error it stays calibrated without refitting, and at 1°/10 cm it lowers the error from 3.93% to 3.39%. With an exposure-robust ZNCC cost, the confidence-based uncertainty breaks down (5.29% vs. 2.80%). Evaluation on real captures with LiDAR ground truth is left to future work and is required before drawing conclusions about real-world accuracy.

1. Introduction

Street-level panoramas are the most widely available imagery of cities. Turning one or two of them into metric 3D geometry that approaches LiDAR quality would make dense 3D maps far cheaper to build and to update. Two families of methods address this today. *Panoramic depth foundation models* [4, 16, 18, 21] predict a depth for every viewing direction from a single panorama, but even the best report relative errors of several percent; at 20 m that is more than a metre. *Generalizable multi-view methods* for panoramas [2, 3, 31] build spherical cost volumes from two or more panoramas and are trained for novel-view synthesis, not for calibrated metric geometry.

A pair of high-resolution panoramas captured a few metres apart carries far more metric information than either family exploits explicitly, but it is distributed very unevenly over the sphere. Triangulation precision degrades as the square of depth and diverges at the *epipoles*, the two directions along the baseline. For a camera on a car or a backpack, the baseline is the direction of travel, so the region without any parallax is precisely the continuation of the street (Fig. 1b). A depth estimator that does not know where geometry is informative must either distrust stereo everywhere or trust it where it is meaningless.

We make this observability explicit and use it for fusion:

- A **closed-form, pose-aware uncertainty** of spherical two-view triangulation for every ray (Sec. 3.1), validated against Monte Carlo simulation within 0.5%, including rotation and translation errors of the relative pose.
- A **geometry-conditioned calibration** (Sec. 3.3): the closed form captures depth, baseline, epipoles, resolution and pose error; only the photometric matching precision is learned, as a two-parameter monotone map fitted robustly on teacher data.
- **Epipole-aware Bayesian fusion** (Sec. 3.4) of a learned panoramic prior with one or more stereo partners in log-depth, with metric scale alignment of the prior and a mixture-model gate against occlusions and moving objects. It outputs metric depth with a calibrated per-ray uncertainty, and reduces exactly to the prior for a single panorama.
- A **city-as-teacher** scheme (Sec. 3.5): a dense capture of one city provides multi-view triangulation covariances for uncertainty-weighted supervision, amodal targets behind occluders seen only from other panoramas, and the cal-

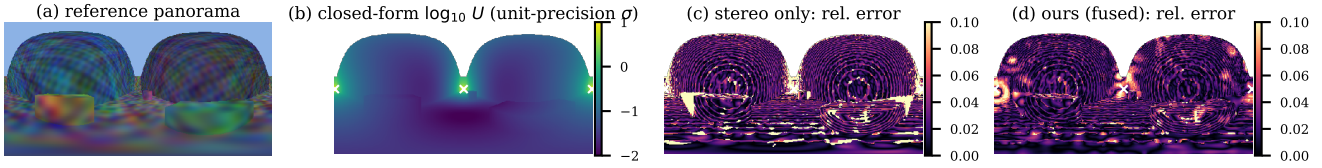


Figure 1. **Two panoramas see depth only away from the epipoles.** (a) Reference panorama of a procedural street; the partner panorama is 4 m further down the street. (b) Our closed-form triangulation uncertainty for unit matching precision: it diverges at the two epipoles ($\times$, looking along the street), exactly where the road continues. (c) Stereo alone fails there. (d) Fusing a monocular panoramic prior with stereo weighted by the calibrated closed-form uncertainty keeps the stereo accuracy where it exists and falls back to the prior where it does not.

ibration data for the stereo precision. We verify the last use; the first two are implemented but not yet evaluated.

- A careful **controlled study** with fitted geometry-blind baselines (Sec. 4). In-distribution, a calibrated match confidence is as good as the closed form, but it breaks down under relative-pose error and with exposure-robust matching costs, where the closed form does not.

We evaluate on a procedural street benchmark with exact ground truth (Sec. 4). We stress that this draft contains *no real-data experiments*; the evaluation protocol for KITTI-360 [15] with accumulated LiDAR is described in Sec. 5 and is the most important missing piece.

2. Related Work

Monocular panoramic depth. Early 360° depth networks fuse equirectangular and cubemap features [10, 14, 24], or merge perspective tangent-image predictions [21]. Recent foundation models predict metric depth for indoor and outdoor panoramas [4, 16] or lift perspective foundation models such as Depth Anything [29] to the sphere [18]. Perspective metric-depth models [1, 19, 30] share the same limitation: a single image constrains metric scale only through learned priors. We treat any such model as a prior with a self-reported uncertainty and correct its scale with geometry.

Multi-view panoramic reconstruction. Omnidirectional stereo with spherical sweeps dates back to OmniMVS [28]. Generalizable radiance-field and Gaussian-splatting methods for wide-baseline panoramas [2, 3, 31] build spherical cost volumes and are trained for view synthesis. Pointmap regressors [13, 25, 26] reconstruct from unposed perspective images. None of these models the per-direction observability of the two-view geometry explicitly or outputs calibrated metric uncertainty.

Triangulation uncertainty. First-order error propagation for triangulation is classical [5, 7], and multi-view stereo pipelines [23] use triangulation angles as heuristics for view selection. We derive the closed form for spherical cameras including relative-pose error, use it as the scale-free backbone of a calibrated uncertainty, and fuse with it rather than threshold it.

Uncertainty in depth estimation. Learned aleatoric uncertainty [11], self-supervised depth uncertainty [20] and sparsification metrics [9] evaluate how well predicted uncertainty ranks errors; calibrated regression [12] recalibrates it post hoc. Our calibration is conditioned on a closed-form geometric quantity that also carries relative-pose error, which a learned confidence cannot observe (Sec. 4.3).

Teacher supervision from more views. Behind the Scenes [27] and its distilled variant [6] train single-view density fields using additional views as supervision on KITTI-360. Our teacher is a metric multi-view reconstruction of a whole city whose per-point covariance weights the loss, and which also supplies amodal targets and the calibration data.

3. Method

Setting. We reconstruct a reference equirectangular panorama I_1 of width W, given a monocular depth prior and optionally one or more partner panoramas I_k with relative poses $(\mathbf{R}_k, \mathbf{b}_k)$ expressed in the reference camera frame (from GNSS/IMU or structure from motion [22]). Every pixel defines a unit ray $\mathbf{d}$; we estimate its metric depth z (distance along $\mathbf{d}$) and an uncertainty. All fusion happens in log-depth $\ell = \log z$, where relative errors are additive.

3.1. Closed-form spherical triangulation uncertainty

Consider one partner at $\mathbf{b}$ and the point $\mathbf{X} = z\mathbf{d}$. In the triangle $(\mathbf{0}, \mathbf{b}, \mathbf{X})$ let α be the angle at the reference camera

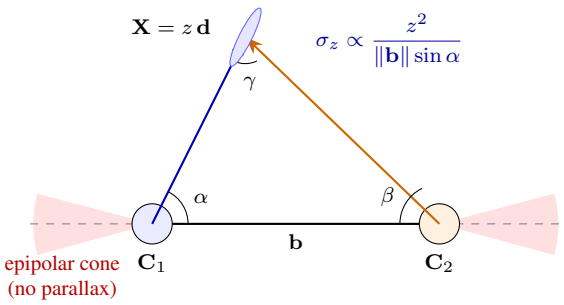


Figure 2. **Spherical two-view geometry.** The depth z along the reference ray $\mathbf{d}$ follows from the law of sines; its uncertainty (Eq. (2)) grows quadratically with depth and diverges inside the epipolar cones along the baseline, which for street-level capture is the direction of travel.

between $\mathbf{d}$ and $\mathbf{b}$, β the angle at the partner, and $\gamma = \pi - \alpha - \beta$ the parallax angle at $\mathbf{X}$. The law of sines gives

$$z = \|\mathbf{b}\| \frac{\sin\beta}{\sin\gamma}. \tag{1}$$

Matching perturbs the two observed directions by small angles within the epipolar plane; out-of-plane components do not change z to first order. Differentiating, $\partial z/\partial\alpha = \|\mathbf{b}\| \sin\beta\cos\gamma/\sin^2\gamma$ and $\partial z/\partial\beta = \|\mathbf{b}\|\sin\alpha/\sin^2\gamma$. A rotation error of the partner with per-axis standard deviation σ_R turns its ray within the epipolar plane by the same angle. A position error with per-axis standard deviation σ_t shifts the partner ray; only its in-plane component perpendicular to that ray matters, and moves the intersection by $\delta/\sin\gamma$. With angular matching precision σ_θ on both rays we obtain

$$\sigma_z^2 = \frac{\|\mathbf{b}\|^2}{\sin^4\gamma}\left[\sigma_\theta^2 \sin^2\beta\cos^2\gamma + (\sigma_\theta^2+\sigma_R^2)\sin^2\alpha\right] + \frac{\sigma_t^2}{\sin^2\gamma}. \tag{2}$$

In the far field ($z \gg \|\mathbf{b}\|$, $\beta \to \pi - \alpha$) and without pose error this reduces to $\sigma_z \approx \sqrt{2}\,\sigma_\theta z^2/(\|\mathbf{b}\|\sin\alpha)$: quadratic in depth, inversely proportional to the baseline, and divergent at the epipoles $\alpha \in \{0, \pi\}$. We compute all sines from cross-product norms; the naive $\sqrt{1-\cos^2}$ loses all precision in single precision for the tiny parallax angles of distant points. A Monte Carlo simulation with isotropic angular noise on both rays and random pose perturbations agrees with Eq. (2) within 0.5% across all tested configurations (unit tests in the code release).

We denote by $U(\mathbf{d}) = \sigma_z/z$ the log-depth uncertainty for *unit* matching precision $\sigma_\theta = 2\pi/W$ (one pixel) without pose error, and by $P(\mathbf{d})$ the pose-only term. U contains all of the geometry: depth, baseline length and direction, and panorama resolution. Because U depends on the unknown depth, we evaluate it at the (scale-aligned) prior depth rather than at the stereo estimate, which is unreliable exactly where U is large.

3.2. Spherical plane-sweep stereo

For each reference ray and log-depth hypothesis we project $z\mathbf{d}$ into the partner panorama and sample it with longitude wrap-around. We sweep 64 log-spaced hypotheses in $[1, 200]$ m, followed by two refinement passes with 16 hypotheses each around the arg-max, and use the soft-arg-min of the final pass. The matching cost is either the box-aggregated L1 difference or $1-$ZNCC over a 5×5 window; ZNCC is invariant to the gain and bias changes between captures. As a match confidence we use the distinctiveness of the coarse cost curve, $c = (\mathrm{med}_h\, C_h - \min_h C_h)/\mathrm{med}_h\, C_h$, which vanishes both near the epipoles, where all hypotheses project to almost the same pixel, and in textureless or occluded regions.

3.3. Geometry-conditioned calibration

Equation (2) is exact *given* σ_θ, but the effective matching precision is not a constant number of pixels. Window aggregation, texture, the cost function and the bounded depth search all matter. On our benchmark, the implied precision varies from 0.1 px near the epipoles to 0.6 px orthogonal to the baseline. We therefore keep the geometry in closed form and calibrate a monotone map from U to the observed error:

$$\sigma_s(\mathbf{d}) = \sqrt{\left(a(c)\,U(\mathbf{d})^\kappa\right)^2 + P(\mathbf{d})^2}, \tag{3}$$

with a shared exponent κ and a scale $a(c)$ that is piecewise constant and non-increasing in the match confidence c (six quantile bins). The pose term P has no free parameters. The parameters are fitted on calibration rays with known depth—in practice the city teacher of Sec. 3.5. We select κ by maximizing a Student-t likelihood ($\nu = 3$), which tolerates occlusions and mismatches. We then fit each a by Gaussian maximum likelihood on the t-inliers, because fusion needs a standard deviation and the t-scale underestimates it by about 20% for Gaussian errors. Rays deep inside the epipolar cones are excluded from fitting: there the bounded depth range truncates the errors. $\kappa = 1$ with a single bin recovers the pure closed form with constant σ_θ.

3.4. Epipole-aware fusion

Scale alignment. Monocular metric depth is often off by a per-image scale factor. We measure a global log-scale offset $\hat{o}$ as the precision-weighted median of $\mu_s - \mu_p$ over well-conditioned rays ($\sigma_s < 0.05$, $c > 0.5$), with weights c/σ_s^2. Its robust standard error is $s_o = 1.2533 \cdot 1.4826\,\mathrm{MAD}/\sqrt{n/25}$; we divide the ray count n by 25 because errors are correlated within the 5×5 aggregation windows. Under a Gaussian prior $\mathcal{N}(0, s_p^2)$ on the scale error of the prior ($s_p = 0.1$), we apply the posterior mean $o = \hat{o}\, s_p^2/(s_p^2 + s_o^2)$. With few or mutually inconsistent reliable rays, as under strong photometric degradation, the correction therefore shrinks towards zero instead of locking onto a wrong scale.

Gated Bayesian fusion. Let the prior be $\mathcal{N}(\mu_p, \sigma_p^2)$ in log-depth and partner k give $\mathcal{N}(\mu_k, \sigma_k^2)$ with confidence c_k. Occlusions, moving objects and mismatches produce gross errors, so we model the disagreement $\delta_k = \mu_k - \mu_p$ as a mixture of an inlier Gaussian $\mathcal{N}(0, \sigma_p^2 + \sigma_k^2)$ with prior probability π and a uniform outlier over the log-depth range L:

$$w_k = c_k \frac{\pi\,\mathcal{N}(\delta_k; 0, \sigma_p^2 + \sigma_k^2)}{\pi\,\mathcal{N}(\delta_k; 0, \sigma_p^2 + \sigma_k^2) + (1-\pi)/L}. \quad (4)$$

The posterior is

$$\frac{1}{\sigma^2} = \frac{1}{\sigma_p^2} + \sum_k \frac{w_k}{\sigma_k^2}, \qquad \mu = \sigma^2\Big(\frac{\mu_p}{\sigma_p^2} + \sum_k \frac{w_k\,\mu_k}{\sigma_k^2}\Big). \quad (5)$$

Near the epipoles $\sigma_k \to \infty$ and the result is the scale-aligned prior; orthogonal to the baseline the stereo dominates. With a single panorama the posterior is the prior. Every operation is differentiable, so the same module can be placed inside a learned network (Sec. 3.5).

3.5. City as teacher

A dense capture of one city, with many panoramas per street, is reconstructed once with multi-view geometry (SfM/MVS [22, 23] or surface splatting [8]). A teacher point $\mathbf{X}$ observed from centres $\mathbf{C}_i$ at distances r_i along unit directions $\mathbf{u}_i$ has triangulation information $\mathbf{\Lambda} = \sum_i (\mathbf{I} - \mathbf{u}_i\mathbf{u}_i^\top)/(r_i\sigma_\theta)^2$, and covariance $\mathbf{\Sigma} = \mathbf{\Lambda}^{-1}$, computed in double precision because single-view points are nearly singular. For a student panorama we render two-layer targets: the first teacher surface along each ray (visible), and the first surface behind it (amodal: behind parked cars, around corners), which only other panoramas of the city observe. Each target carries the variance $\mathbf{d}^\top\mathbf{\Sigma}\mathbf{d}$ along the student ray. A student predicting (μ, σ) in log-depth is trained with the Gaussian negative log-likelihood whose variance is $\sigma^2 + \mathbf{d}^\top\mathbf{\Sigma}\mathbf{d}/z^2$, so poorly triangulated teacher points supervise weakly instead of wrongly. The same teacher provides the calibration rays for Eq. (3). Our reference student shares an equirectangular encoder (with longitude wrap-around padding) between a prior head, the stereo cost volume on learned features, the closed-form uncertainty with a learned global scale, the fusion of Sec. 3.4, and an amodal second-layer head.

4. Experiments

4.1. Setup

Benchmark. Real street panoramas with dense metric ground truth are not available to us at the time of writing (Sec. 5). We therefore use procedural street canyons with exact depth: a ground plane, two building façades 5–9 m from the camera with random heights, end walls 60–120 m away, 3–7 box-shaped parked vehicles, and sky. All surfaces carry a view-independent multi-scale procedural texture. The reference panorama is at the origin and the partner b metres further along the street, the hardest configuration, because the epipoles look down the street. Unless stated otherwise, panoramas are 256×512 and $b = 4$ m. The monocular prior is simulated as the true log-depth plus spatially smooth noise with 6% standard deviation and a global scale error drawn uniformly in $\pm 10\%$, with a constant self-reported $\sigma_p = 0.06$; this matches the reported accuracy of current panoramic foundation models [16]. Calibration uses six scenes (seeds ≥ 1000), disjoint from the ten test scenes, with ground truth standing in for the city teacher.

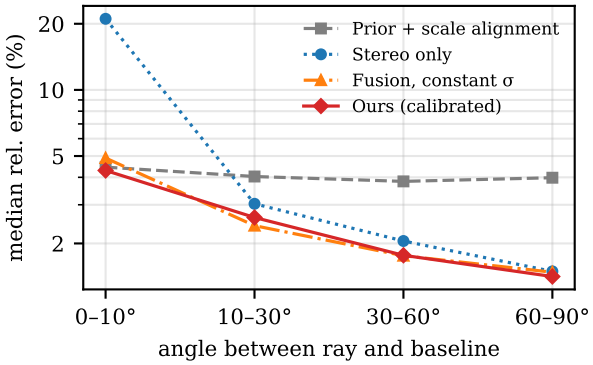


Figure 3. Median relative error versus the angle between ray and baseline. Stereo diverges near the epipoles; the scale-aligned prior is flat; fusion follows the lower envelope.

Baselines. All fusion variants share the same stereo, prior, scale alignment and gate, and differ only in the stereo uncertainty σ_s: *constant* ($\sigma_s = s$) and *confidence* ($\sigma_s = s/c$), with s fitted on the same calibration rays and with the same robust estimator as ours; *selection*, which takes per ray the source with the smaller calibrated σ; and our closed form without calibration ($\sigma_\theta = 0.5$ px). We also report the prior alone, the scale-aligned prior and stereo alone.

Metrics. Median relative depth error overall and per bin of the angle between ray and baseline; the fraction of rays within 20 cm; the fraction of errors within the predicted 1σ (ideal 0.683); the Gaussian negative log-likelihood (NLL) in log-depth, a proper scoring rule for the predicted distribution; and the area under the sparsification error curve (AUSE) [9].

4.2. Main results

Table 1 and Fig. 3 show the expected structure. Stereo alone is accurate orthogonal to the baseline (1.49% at 60–90°) but useless near the epipoles (21%). The prior is uniformly

	Median rel. error (%) ↓, by angle to baseline					<20 cm (%) ↑	Cal. 1σ	NLL ↓	AUSE ↓
Method	0–10°	10–30°	30–60°	60–90°	all	all	(0.68)	all	all
Prior only	6.81	7.04	6.99	6.92	6.95	29.80	0.44	-0.73	0.481
Prior + scale alignment	4.45	4.03	3.83	3.98	3.94	47.12	0.69	-1.40	0.534
Stereo only	21.07	3.03	2.05	1.49	1.78	68.99	**0.68**	26.04	0.386
Fusion, constant σ_s	4.90	2.41	1.76	1.48	1.64	73.66	0.73	-2.17	**0.329**
Fusion, confidence σ_s	4.49	**2.35**	**1.73**	1.43	**1.60**	**74.10**	0.73	**-2.22**	0.331
Selection (min. σ)	4.45	3.07	2.03	1.47	1.74	70.60	0.70	-2.03	0.354
Ours, closed form (σ_{px}=0.5)	4.38	3.03	1.91	1.43	1.67	72.17	0.63	-1.36	0.461
Ours (calibrated)	**4.30**	2.63	1.77	**1.41**	1.61	73.63	0.71	-2.16	0.361

Table 1. **Main results** (256×512, $b = 4$ m along the street, 10 test scenes). The calibrated closed form is best or tied in accuracy near the epipoles and orthogonal to the baseline, and is well calibrated. A geometry-blind confidence-based σ_s, fitted on the same calibration data, is equally accurate in this in-distribution setting (see text).

mediocre, 6.95% before and 3.94% after scale alignment. Our fusion follows the lower envelope of the two, reducing the median error of the raw prior from 6.95% to 1.61% overall while remaining calibrated (0.71 of errors within 1σ). Scale alignment matters: without it the error near the epipoles rises from 4.30% to 6.51%. Evaluating U at the stereo depth is equivalent once the precision is calibrated. The uncalibrated closed form ($\sigma_\theta = 0.5$ px) is as accurate overall but badly calibrated (NLL -1.36 vs. -2.16), which motivates Eq. (3): on this benchmark, the implied matching precision varies by $6\times$ with the angle to the baseline.

An honest negative result. With its scale fitted on the same calibration rays, the geometry-blind *confidence* baseline is as accurate as ours in-distribution (1.60% vs. 1.61%) and has a slightly better NLL. The match confidence already collapses near the epipoles, because all hypotheses project to nearly the same pixel, so it implicitly encodes much of the observability. The same holds when transferring the calibration to other baselines and resolutions (Tab. 2): all calibrated fusion variants are within 0.1 percentage points of each other. The closed form therefore does *not* improve accuracy over a well-calibrated confidence in the absence of pose error on this benchmark.

4.3. Where geometry matters: relative-pose error

Poses from GNSS/IMU or SfM are never exact, and their error affects rays very differently: by Eq. (2), a translation error moves the depth by $\delta / \sin\gamma$, which is negligible for close points orthogonal to the baseline and dominant for distant points and near the epipoles. Match confidence cannot see this, because the cost curve can be sharply peaked at a wrong depth. Table 3 injects pose errors into the partner pose. For small errors (0.1°, 2 cm) all methods are equivalent. At 0.3°/5 cm the geometry-blind uncertainties become over-confident (0.59–0.60 of errors within 1σ), while ours stays calibrated (0.70) and has the best NLL, at equal accuracy. At 1°/10 cm the pose term changes the result: ours reduces the median error from 3.93% (best baseline) to 3.39%, raises the in-1σ fraction from 0.46 to 0.67, and improves the NLL from -0.82 to -1.54. Refitting the geometry-blind baselines under the same pose noise restores calibration at moderate noise. At large noise, however, it inflates σ_s uniformly, so no ray remains reliable enough for scale alignment and the result degrades towards the unaligned prior (6.5–6.8%). Ours keeps rays orthogonal to the baseline at close range trustworthy, because there the pose term is small. The pose term assumes the noise level is known; in practice it would come from the SfM covariance or the GNSS/IMU specification.

Partners with different baselines. With two partners at +2 m and −8 m, all calibrated variants reach the same accuracy (1.19–1.21%) and similar calibration, *i.e.* we observe no benefit of geometric weighting across partners on this benchmark.

4.4. Capture effects

Table 4 varies the capture conditions. Two findings stand out. First, the cost function matters: L1 is best for clean images and under sensor noise, while ZNCC is far more robust to an exposure change between captures (2.88% vs. 5.96%) and in the combined setting (4.56% vs. 8.51%). Second, with ZNCC the geometry-blind confidence baseline breaks down (5.29% vs. 2.80% on clean data, 6.72% vs. 4.56% combined). Its success with L1 rests on the L1 confidence happening to encode the epipolar degeneracy, which does not hold for every cost. The closed form does not depend on such a coincidence. The outlier gate gives a modest gain on clean data and with moving cars (3.06%→2.80% and 3.09%→2.82%), and none in the combined setting.

Method	b=2 m		b=8 m		128×256		512×1024	
	Med. (%)	NLL	Med. (%)	NLL	Med. (%)	NLL	Med. (%)	NLL
Prior + scale alignment	3.92	-1.39	3.91	-1.39	3.99	-1.37	3.97	-1.39
Stereo only	1.65	2.02	2.16	133.00	2.21	5.59	1.49	75.30
Fusion, constant σ_s	1.57	-2.25	1.82	-1.87	1.94	-2.03	1.46	-2.18
Fusion, confidence σ_s	1.52	-2.32	1.81	-1.92	1.89	-2.10	1.41	-2.22
Ours (calibrated)	1.51	-2.29	1.82	-1.89	1.97	-2.03	1.38	-2.00

Table 2. **Transfer without recalibration**: all methods are calibrated at $b = 4$ m and 256×512, and evaluated at other baselines and resolutions (5 scenes; 3 at 512×1024).

Method	0.1°/2cm			0.3°/5cm			1°/10cm		
	Med.	Cal.	NLL	Med.	Cal.	NLL	Med.	Cal.	NLL
Constant σ_s	1.77	0.70	-2.08	2.34	0.59	-1.70	3.99	0.44	-0.51
Constant σ_s, refit	1.79	0.73	-2.09	2.41	0.71	-1.83	6.77	0.44	-0.77
Confidence σ_s	**1.73**	0.70	-2.13	2.30	0.60	-1.79	3.93	0.46	-0.68
Confidence σ_s, refit	1.74	0.72	**-2.13**	2.36	0.71	-1.87	6.52	0.44	-0.82
Ours w/o pose term	1.74	**0.69**	-2.08	**2.26**	0.60	-1.78	3.86	0.46	-0.68
Ours	1.75	0.71	-2.09	2.28	**0.70**	**-1.89**	**3.39**	**0.67**	**-1.54**

Table 3. **Relative-pose error.** The partner pose given to all methods is perturbed by rotation/translation noise with the stated per-axis standard deviations; images are rendered with the true poses. "refit": geometry-blind baselines recalibrated on calibration scenes with the same pose noise. Ours uses the known noise level in the closed-form pose term (Eq. (2)) and is *not* recalibrated. 8 test scenes.

Method	clean	exposure	noise	moving cars	all
Confidence σ_s, L1	**1.59**	5.69	2.26	**1.62**	9.07
Confidence σ_s, ZNCC	5.29	5.42	6.50	5.32	6.72
Ours, L1	1.60	5.96	**2.24**	1.63	8.51
Ours, ZNCC, w/o outlier gate	3.06	3.06	3.95	3.09	4.57
Ours, ZNCC	2.80	**2.88**	3.92	2.82	**4.56**

Table 4. **Capture effects** (median relative error, %, 8 scenes): partner exposure change (gain 0.6, bias 0.15), sensor noise ($\sigma = 0.03$), half of the parked cars moved by 3 m between captures, and all three combined. Each cost function has its own calibration.

Calibration data	κ	Med. rel. (%)	Cal. 1σ	NLL
Ground truth (oracle)	0.38	1.61	0.71	-2.16
Teacher, naive	0.40	1.62	0.73	-2.15
Teacher, $\geq$2 views	0.35	1.62	0.73	-2.16
Teacher, covariance-aware (ours)	0.30	1.61	0.73	-2.17

Table 5. **Calibration data.** The precision model of Eq. (3) is fitted on depth from a simulated city teacher (7 panoramas per street, lateral offsets, points displaced by draws from their own triangulation covariance, 0.5 px matching at 2048 px width) instead of ground truth. Evaluation as in Tab. 1.

4.5. Calibrating from a noisy city teacher

Fitting the calibration on noisy teacher depth instead of ground truth costs nothing measurable (Tab. 5): all variants reach 1.61–1.62% with calibrated uncertainties. Treating the teacher as exact inflates the fitted scales slightly, as expected, because teacher error is absorbed into the matching precision. Adding the teacher covariance in quadrature removes this inflation. Downstream, however, the difference is negligible here, because the simulated teacher is about $8\times$ more precise than the 512 px stereo it calibrates. Covariance-aware calibration should matter only for teachers of comparable precision; we have not tested that regime.

5. Limitations and Future Work

No real data yet. All results in this draft are on procedural scenes with a simulated prior. Real panoramas add non-Lambertian surfaces, vegetation, sky, rolling-shutter and stitching artefacts, and a prior whose errors are spatially structured in ways our simulation does not capture. The decisive experiment is KITTI-360 [15]: its two 180° fisheye cameras (MEI model [17]) cover nearly the full sphere and can be resampled to equirectangular images, and accumulated LiDAR provides metric ground truth including occluded regions. The protocol is: pairs of frames along the trajectory at 2–8 m spacing, priors from Depth Any Panoramas [16] and DA360 [4], calibration on training sequences, and evaluation on held-out sequences against Splatter-360 [3], PanSplat [31] and pointmap regressors [13, 25] run on cube faces.

The learned student is not evaluated. The city-as-teacher training (Sec. 3.5) is implemented. A small reference student trained for 400 CPU steps did not converge (median errors far above the prior), so we report no student results. The teacher is evaluated only as a source of calibration data (Sec. 4.5). Whether amodal supervision and covariance-weighted teacher losses improve a foundation-model student requires GPU training and remains open.

Modelling assumptions. Partners share the reference image, so treating several partners as independent overstates the precision. Moving objects are rejected as outliers, and their depth falls back to the prior. Relative-pose error enters σ_s only through a per-axis standard deviation. When poses come from SfM, their actual covariance should be propagated instead.

6. Conclusion

Two panoramas carry precise metric information about a scene, but only away from their epipoles. We showed that a closed-form, pose-aware triangulation uncertainty, combined with a small geometry-conditioned calibration of the matching precision, makes this observability explicit. Fusing on that basis with a monocular prior yields metric depth with calibrated per-ray uncertainty that falls back gracefully where geometry is uninformative. In controlled experiments it reduces the error of a simulated foundation prior by more than $4\times$. In-distribution it is matched by a well-calibrated geometry-blind confidence. It remains calibrated and becomes clearly more accurate once the relative pose is uncertain, the normal case for street-level capture, and when an exposure-robust cost is used, where the confidence proxy loses its implicit knowledge of the epipoles. Establishing how close this brings real street-level panoramas to LiDAR accuracy is the next step.

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